

Task-1& 3 (Ritesh)

Task-1

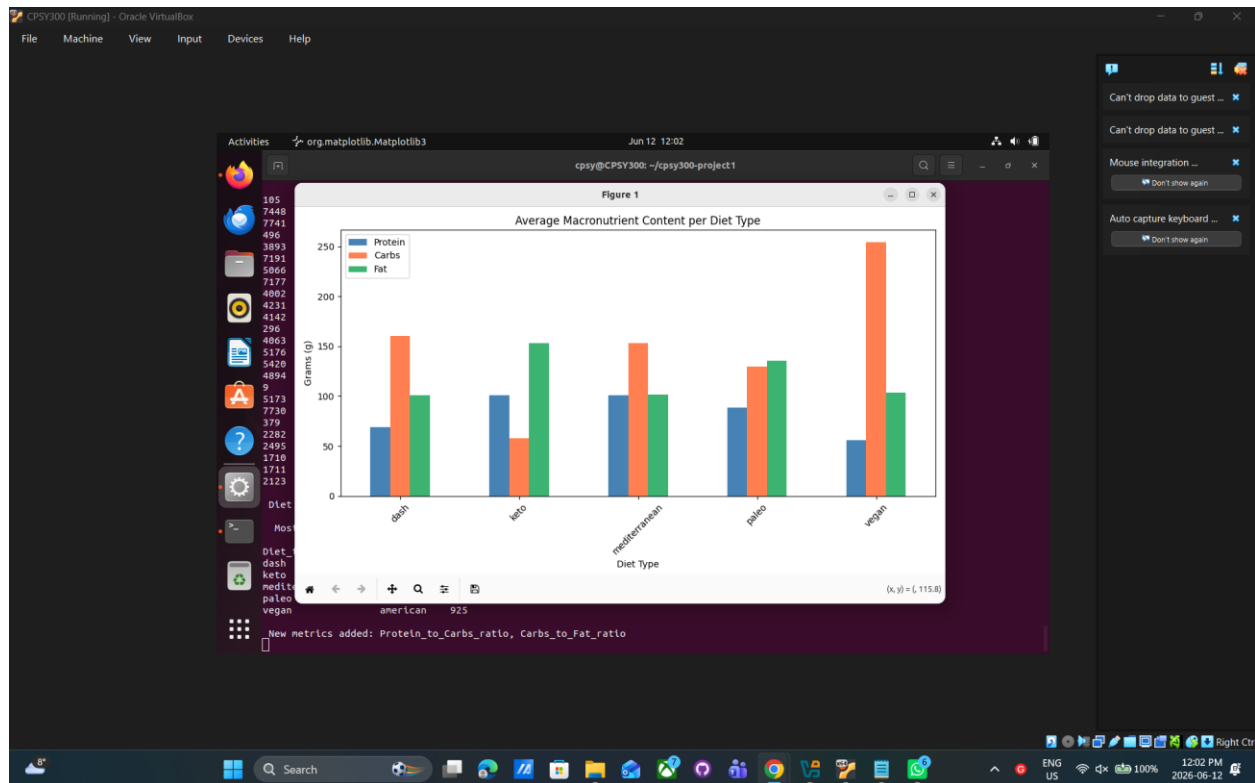


Figure 1: Bar Chart – Average Macronutrient Content (Protein, Carbs, Fat) per Diet Type generated using Matplotlib.

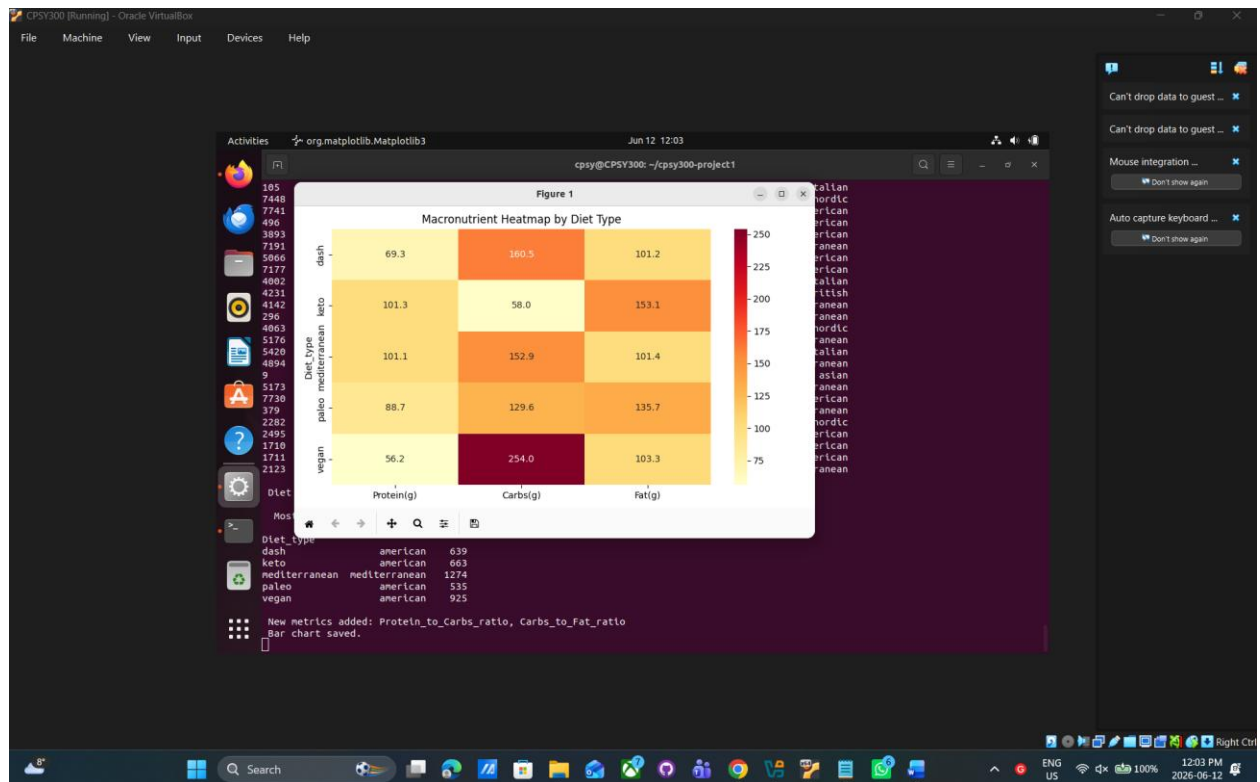


Figure 2: Heatmap – Macronutrient content by Diet Type showing Vegan has highest Carbs and Keto has lowest Carbs, generated using Seaborn.

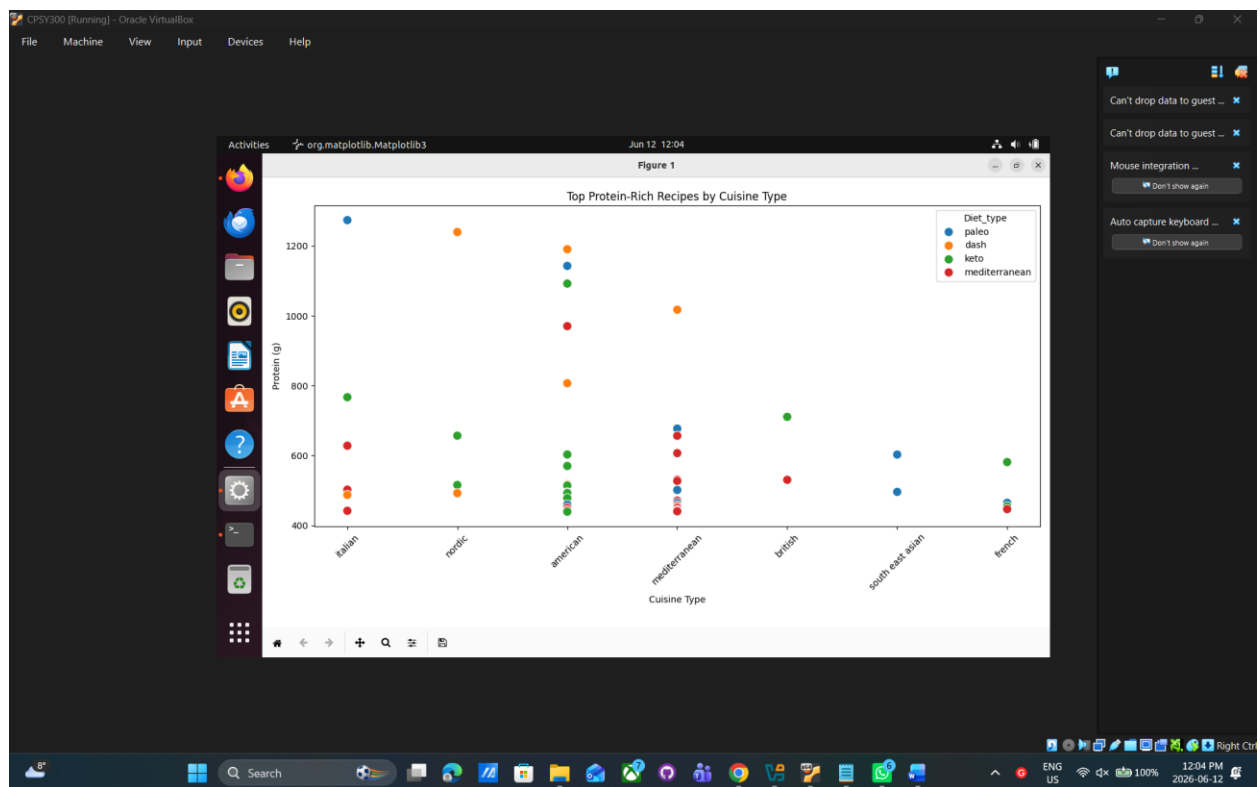


Figure 3: Scatter Plot – Top Protein-Rich Recipes distributed across Cuisine Types and Diet Types.

```
cpsh@CPSY300: ~/cpsy300-project1
3893 keto Sara Louise's Keto Smoked Holiday Turkey 1892.00 american
7191 dash Barbecue Chicken Legs 1817.25 mediterranean
5966 mediterranean Fava Bean Salad with Mountain Ham and Mint 970.31 american
7177 dash 12th Man Hot Wings 807.03 american
4002 keto Mayo Free Deviled Eggs (Paleo, Whole30 + Keto) 766.99 italian
4231 keto Low Carb Beef and Cheddar Cauliflower Bake, TH... 710.81 british
4142 keto Orange and Five-Spice Roasted Chicken Legs | K... 677.22 mediterranean
296 paleo Orange and Five-Spice Roasted Chicken Legs | K... 677.22 mediterranean
4063 keto Low Carb Salmon Cakes (Keto, Grain Free) 656.95 nordic
5176 mediterranean Poached Salt Cod with Vegetables and Garlic Mo... 656.65 mediterranean
5420 mediterranean Mediterranean Pizza 628.29 italian
4894 mediterranean Orange and Cumin Leg of Lamb with Roasted Tona... 607.83 mediterranean
9 paleo Vietnamese Pho Pressure Cooker (Noodle Soup) 602.91 south east asian
5173 mediterranean Meatballs Braised with Kale 531.48 mediterranean
7730 dash Jazzed Up Turkey Tetrazzini 517.47 american
379 paleo Paleo Slow Cooker Honey Garlic Chicken and Veg... 501.65 mediterranean
2282 vegan Tanya Tertyaki Salmon 431.10 nordic
2495 vegan Mint nut roasts with candied carrots 421.72 american
1710 vegan Vegan Pinto Bean Chili 381.55 american
1711 vegan Vegan Chili and Cornbread Casserole 319.54 american
2123 keto Kentucky fried seitan 317.43 mediterranean

Diet with highest average protein: keto

Most Common Cuisine per Diet Type:
Diet_type cuisine_type count
dash american 639
keto american 663
mediterranean mediterranean 1274
paleo american 535
vegan american 925

New metrics added: Protein_to_Carbs_ratio, Carbs_to_Fat_ratio
Bar chart saved.
Heatmap saved.
Scatter plot saved.

Task 1 complete! All visualizations saved to /visualizations/
cpsh@CPSY300: ~/cpsy300-project1$
```

Figure 4: Terminal Output – data_analysis.py executed successfully showing average macronutrients, top protein recipes, most common cuisines, new metrics added, and all 3 visualizations saved.

Task-3

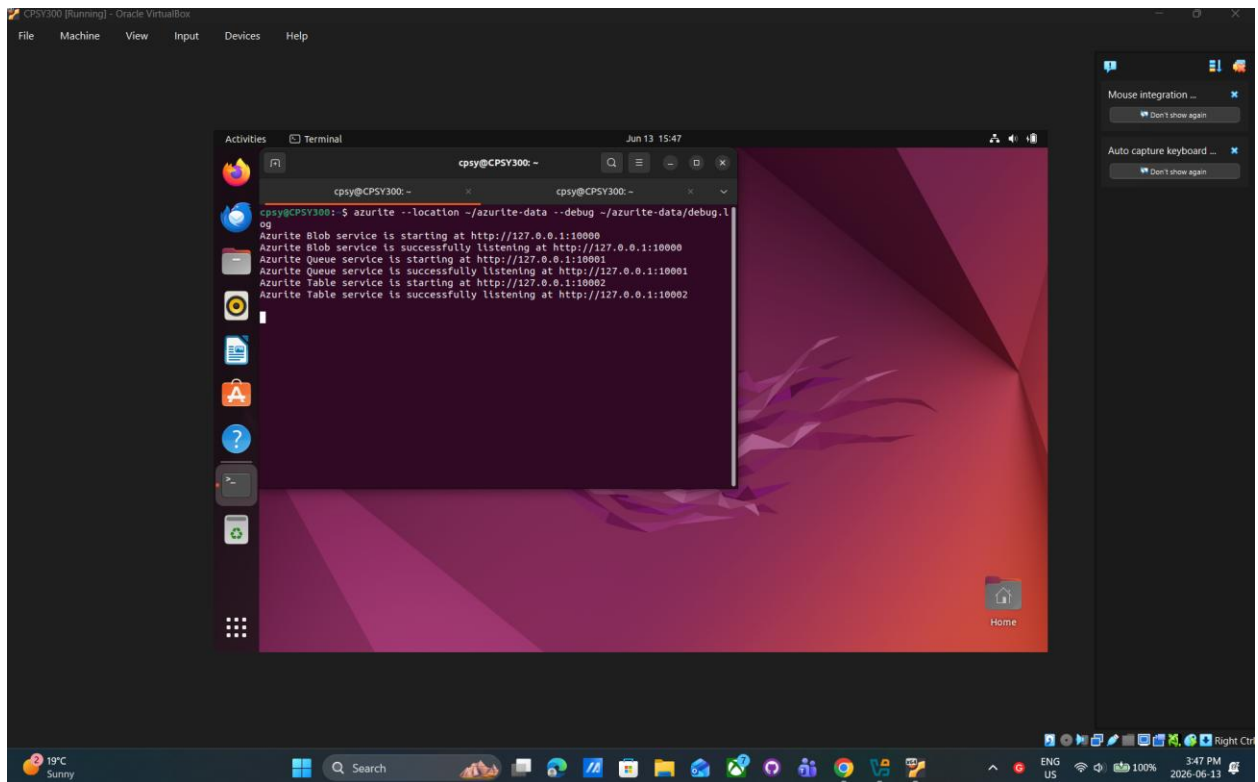
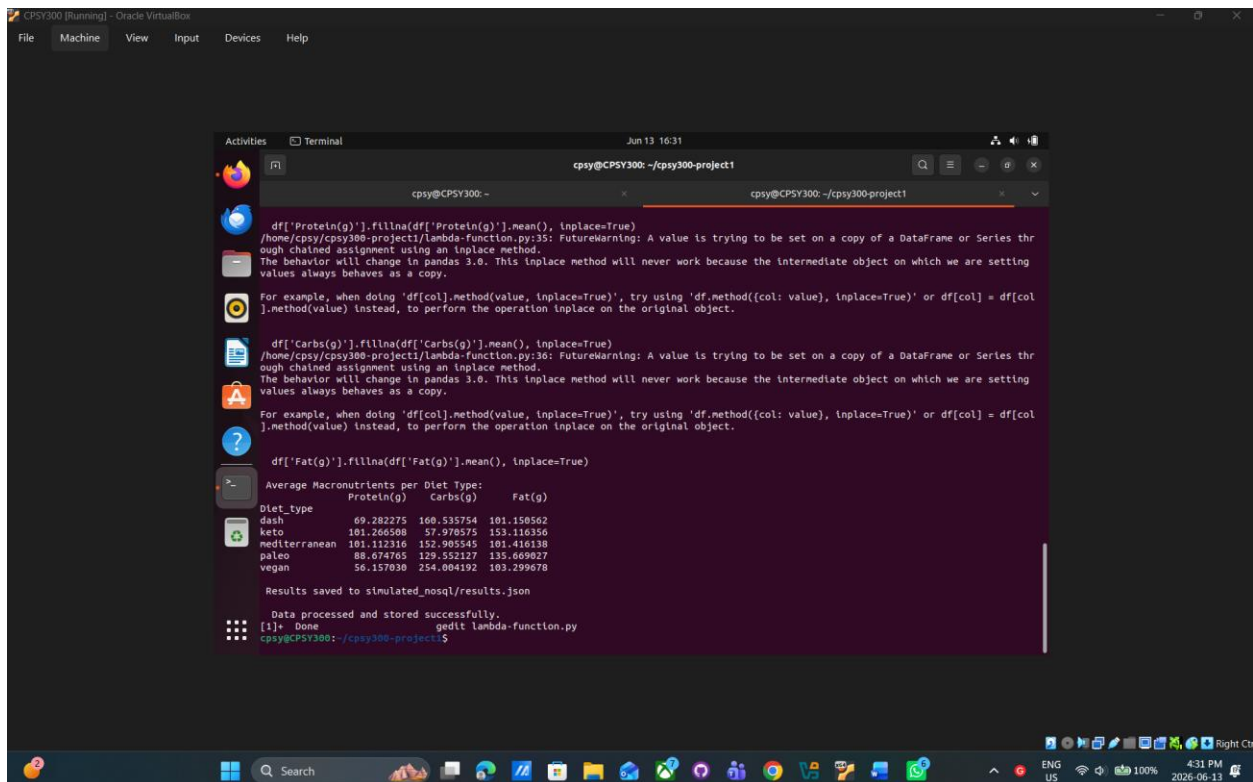


Figure 5: Azurite Local Blob Storage Emulator running successfully with Blob service on port 10000, Queue service on port 10001, and Table service on port 10002 — simulating Azure cloud storage locally.



```
df['Protein(g)'].fillna(df['Protein(g)'].mean(), inplace=True)
/home/cpsy/cpsy300-project1/lambda-function.py:35: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

df['Carbs(g)'].fillna(df['Carbs(g)'].mean(), inplace=True)
/home/cpsy/cpsy300-project1/lambda-function.py:36: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

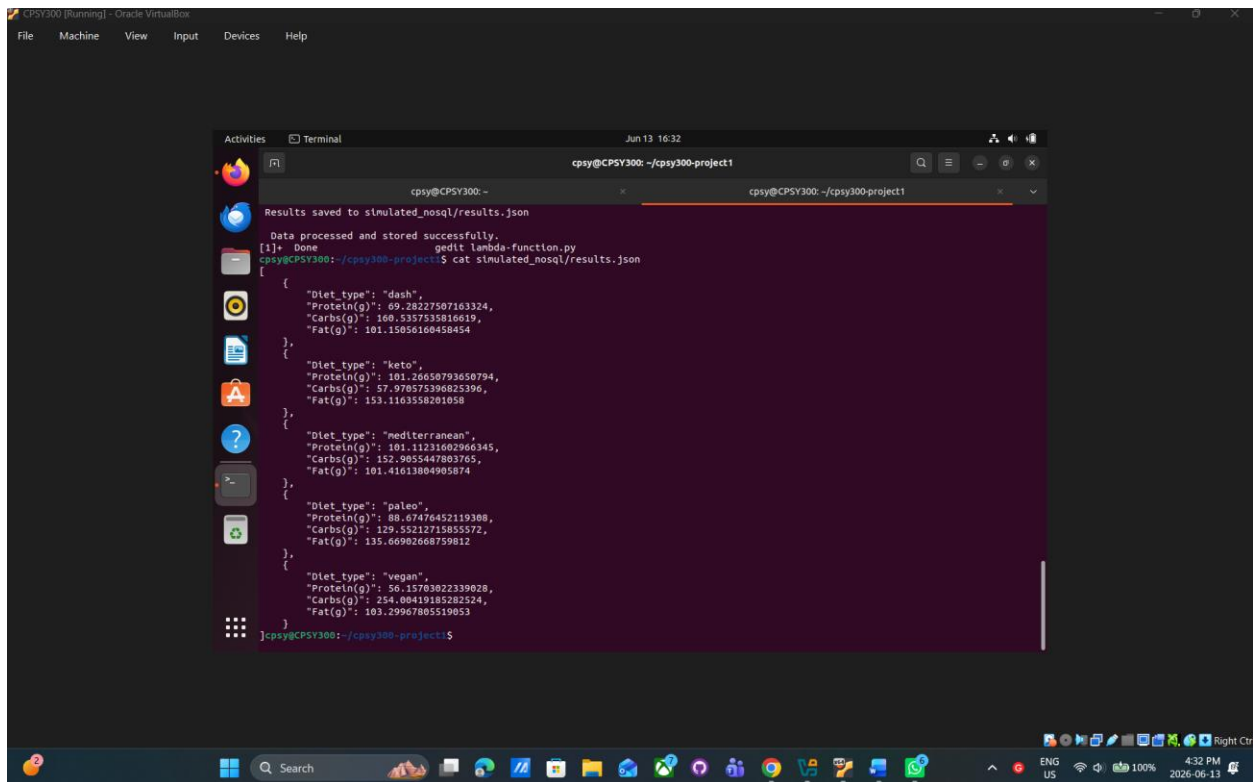
df['Fat(g)'].fillna(df['Fat(g)'].mean(), inplace=True)

Average Macronutrients per Diet Type:
  Diet_Type  Protein(g)  Carbs(g)  Fat(g)
dash      69.282275    160.525754  101.150562
keto      101.266508    57.970575   153.116356
mediterranean 101.112316  152.905545  101.416138
paleo      88.074765    129.552127  135.669827
vegan      56.157030    254.004192  103.299678

Results saved to simulated_nosql/results.json

Data processed and stored successfully.
[1]+  Done                  gedit lambda-function.py
cpsh@CPSY300: ~/cpsh300-project1$
```

Figure 6: lambda_function.py executed successfully — connected to Azurite, uploaded All_Diets.csv, processed data, and calculated average macronutrients per diet type. Results saved to simulated_nosql/results.json.



```
Results saved to simulated_nosql/results.json
Data processed and stored successfully.
[1]+  Done                  gedit lambda-function.py
cpsy@CPSY300: ~/cpsy300-project1$ cat simulated_nosql/results.json
[
  {
    "Diet_type": "dash",
    "Protein(g)": 69.28227507163324,
    "Carbs(g)": 160.5357535816619,
    "Fat(g)": 101.15056160458454
  },
  {
    "Diet_type": "keto",
    "Protein(g)": 101.26650793650794,
    "Carbs(g)": 57.970575396825396,
    "Fat(g)": 153.1163558261058
  },
  {
    "Diet_type": "mediterranean",
    "Protein(g)": 101.11231602966345,
    "Carbs(g)": 152.9055447803765,
    "Fat(g)": 101.41613804905374
  },
  {
    "Diet_type": "paleo",
    "Protein(g)": 88.67476452119308,
    "Carbs(g)": 129.55212715855572,
    "Fat(g)": 135.66902668759812
  },
  {
    "Diet_type": "vegan",
    "Protein(g)": 56.15703022339028,
    "Carbs(g)": 254.00419185282524,
    "Fat(g)": 103.29967805519053
  }
]
```

Figure 7: results.json file contents showing simulated NoSQL database output with average Protein, Carbs, and Fat values for all 5 diet types (dash, keto, mediterranean, paleo, vegan).